



Name a Martian Rover

The European Space Agency has opened up the naming of its ExoMars rover to the public. <https://digit.in/Aug18-61-1>



Crystallised Magma on Mars

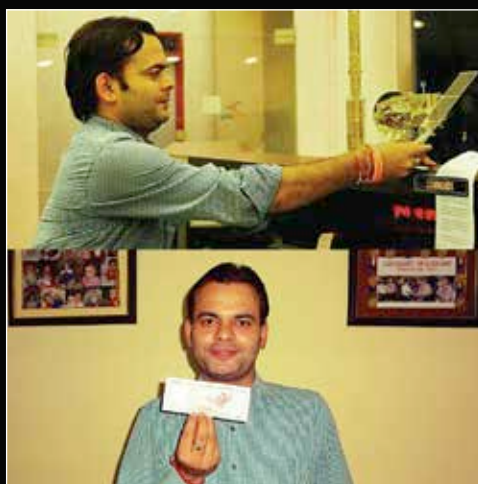
A new study indicates that the crust of Mars could be made out of crystallised Magma. <https://digit.in/Aug18-61-2>

Over twenty scientific papers have been published in peer reviewed journals, based on the data gathered by the Mangalyaan mission. The data captured by the mission is available through the Indian Space Science Data Center (ISSDC) web site. The archived data is available for free. In the first year that the data was made available to the public, the site saw over 175,000 hits, with over 200 Gb of data being downloaded. Since science was not the primary objective of the mission, we asked Sharma exactly how the data is proving to be useful. "Mangalyaan has given a huge amount of science data. There are many scientific organizations and scientists working in this field who don't have actual and current data. The data may help them establish some science goal or retune their models."

One of the successes of the scientific study of the mission, is that Mangalyaan is not in a circular orbit around the planet. Instead, the spacecraft is in an eccentric or elliptical orbit. The orbit is similar to the way a comet orbits around the Sun. At its closest point to Mars, known as the periareon, Mangalyaan is at a distance of 421 kilometers. At its farthest point, at its apoareon, the spacecraft is at a distance of a distance of 76,993 kilometers. This unique orbit allows the MoM to capture data that is not possible by any of the other contemporary Mars orbiters. We asked Sharma why this orbit was chosen for Mangalyaan, and what are the benefits of having done so, "based on the capabilities of the PSLV-XL and the position of the Sun in the Mars phase of the mission, the orbit of the spacecraft was decided to be elliptical. This orbit is unique and gives us many firsts, like a full disk image of Mars in one snap, and an image of the far side Phobos." The unique orbit allows the mission to study the moons of Mars from unique angles, including the moons with Mars in the background. "Mangalyaan has captured many images of Phobos and Deimos. Some of the unique images included

a rising Phobos and the far side of Deimos." A full disk image means a single capture where the entire planet is visible. The November 2016 issue of the National Geographic magazine featured one such global disc image of Mars captured by MoM. Over 900 images have been beamed back by the mission so far. A number of images captured by the mission has been compiled in an "Atlas of Mars" by ISRO (<https://digit.in/momatlas>).

A key to extending the life of the mission was handling an event known as the Mars Solar Conjunction. About once every 26 months, the Earth, the Sun and Mars align along a single line.



KK Sharma at the command console, and with the new ₹2000 note

This means that the Sun blocks out communication between Earth and Mars. During the communications blackout, ISRO would not be able to send and receive signals to and from the orbiter, because of charged particles from the Sun introducing an excessive amount of noise to the signals. The spacecraft had to manage itself entirely autonomously during this period. ISRO ran simulations to ensure that the spacecraft would survive this period of communications blackout. We asked Sharma how the situation on the ground was during the communications blackout. "During this period there was no communication possible. Before starting of these periods all possible autonomy

of the satellite was enabled to take care of any undesired situation. Since during this time there is no telemetry we have to rely on autonomy. After the Sun moved away, we started receiving data and analysed the satellite's condition. No undesired things happened during this blackout period and satellite was in good condition as expected." We then asked exactly how the autonomy on the spacecraft worked. "Due to two way communication delay time, which is variable, we were relying on the autonomous features of the spacecraft. For autonomy, we consider all possible failure cases in advance and based on the failure conditions, the solutions are also pre programmed. To take care of this, the spacecraft has main and redundant systems for almost everything. So, whenever a failure and non nominal condition is detected by the spacecraft, the autonomy triggers and brings the satellite back to nominal."

Another challenge to the longevity of the mission was to ensure that the solar panels received enough light from the Sun to keep the batteries on board powered. If Mars came in between the orbiter and the Sun for more than a period of 8 hours, then the instruments on board would no longer be able to function. In January 2017, ISRO executed an orbital correction maneuver, that ensured that the spacecraft would not experience another long period eclipse till at least the year 2021.

ISRO is planning a follow up mission to Mars, Mangalyaan-2. The second mission will also be an orbiter, with an elliptical orbit, and will have additional scientific payloads on board. Science will be the primary objective of the second Indian mission to Mars. It is very likely that the first will still be in orbit around Mars, continuing to send back important data that will help us increase our understanding of the Red Planet. 